INSTITUTO TECNOLÓGICO Y DE ESTUDIOS SUPERIORES DE MONTERREY

Actividad 6.2. Ejercicio de programación 3

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Pruebas de software y aseguramiento de la calidad **22 de febrero del 2026**

Hotel Test Results

```
(base) PS C:\Users\delrivero.m> & C:/Users/delrivero.m/AppData/Local/Programs/Python/Python313/python.exe c:/Users/delrivero.m/a01362368_NumerodeActividadA6.2/tests/test_hotel.py
[ERROR] Hotel 'H1' already exists.
..[ERROR] Hotel 'GHOST' not found.
..--- Hotel ---
  hotel_id: H3
  name: Mountain Lodge
  location: Denver
  total_rooms: 15
  available_rooms: 15
  reservations: []
.[ERROR] Hotel 'GHOST' not found.
..[ERROR] Hotel 'GHOST' not found.
.[WARN] Field 'color' is not modifiable or unknown.
.[ERROR] Failed to load hotels data: Expecting value: line 1 column 1 (char 0)
..[ERROR] Failed to save hotels data: disk full
.[ERROR] Hotel 'GHOST' not found.
.[ERROR] Reservation 'GHOST_RES' not in hotel.
..[ERROR] Hotel 'GHOST' not found.
.[ERROR] No available rooms in hotel 'R1'.
```

Customer Test Results

```
(base) PS C:\Users\delrivero.m> & C:/Users/delrivero.m/AppData/Local/Programs/Python/Python313/python.exe c:/Users/delrivero.m/a01362368_NumerodeActividadA6.2/tests/test_customer.py
[ERROR] Customer 'C1' already exists.
..[ERROR] Customer 'GHOST' not found.
..--- Customer ---
  customer_id: C3
  name: Carol
  email: carol@x.com
  phone: 555-9999
.[ERROR] Customer 'GHOST' not found.
..[ERROR] Customer 'GHOST' not found.
.[WARN] Field 'age' is not modifiable or unknown.
.[ERROR] Failed to load customers data: Expecting value: line 1 column 1 (char 0)
..[ERROR] Failed to save customers data: no space
.
----------------------------------------------------------------------
Ran 12 tests in 0.022s

OK
```

Reservations Test Results

```
OK
(base) PS C:\Users\delrivero.m> & C:/Users/delrivero.m/AppData/Local/Programs/Python/Python313/python.exe c:/Users/delrivero.m/a01362368_NumerodeActividadA6.2/tests/test_reservation.py
[ERROR] Reservation 'R4' already cancelled.
.[ERROR] Reservation 'GHOST' not found.
..[ERROR] Reservation 'R1' already exists.
.[ERROR] Hotel 'GHOST' not found.
.[ERROR] No available rooms in hotel 'H_SMALL'.
..[ERROR] Failed to load reservations data: Expecting value: line 1 column 1 (char 0)
..[ERROR] Failed to save reservations data: disk full
.
----------------------------------------------------------------------
Ran 10 tests in 0.038s
```

Flake8 results

First Run

```
(base) PS C:\Users\delrivero.m\a01362368_NumerodeActividadA6.2> python -m flake8 models/ tests/ main.py --max-line-lengt
h=99
models/hotel.py:105:15: F541 f-string is missing placeholders
models/reservation.py:5:1: F401 'datetime.date' imported but unused
models/reservation.py:118:1: W293 blank line contains whitespace
models/reservation.py:118:5: W292 no newline at end of file
tests/test_customer.py:6:1: E402 module level import not at top of file
tests/test_customer.py:7:1: E402 module level import not at top of file
tests/test_customer.py:9:1: E402 module level import not at top of file
tests/test_customer.py:10:1: E402 module level import not at top of file
tests/test_hotel.py:3:1: F401 'json' imported but unused
tests/test_hotel.py:7:1: E402 module level import not at top of file
tests/test_hotel.py:8:1: E402 module level import not at top of file
tests/test_hotel.py:10:1: E402 module level import not at top of file
tests/test_hotel.py:11:1: E402 module level import not at top of file
tests/test_reservation.py:6:1: E402 module level import not at top of file
tests/test_reservation.py:7:1: E402 module level import not at top of file
tests/test_reservation.py:9:1: E402 module level import not at top of file
tests/test_reservation.py:10:1: E402 module level import not at top of file
tests/test_reservation.py:11:1: E402 module level import not at top of file
tests/test_reservation.py:12:1: E402 module level import not at top of file
(base) PS C:\Users\delrivero.m\a01362368_NumerodeActividadA6.2> git status
On branch main
Your branch is up to date with 'origin/main'.
```

End Result

```
(base) PS C:\Users\delrivero.m\a01362368_NumerodeActividadA6.2> python -m flake8 models/ tests/ main.py --max-line-lengt
h=99
(base) PS C:\Users\delrivero.m\a01362368_NumerodeActividadA6.2>
```

Pylint results

First Run

```
(base) PS C:\Users\delrivero.m\a01362368_NumerodeActividadA6.2> python -m pylint models/ tests/ main.py
************* Module models.reservation
models\reservation.py:35:4: R0913: Too many arguments (6/5) (too-many-arguments)
models\reservation.py:35:4: R0917: Too many positional arguments (6/5) (too-many-positional-arguments)
models\reservation.py:75:8: C0415: Import outside toplevel (models.hotel.Hotel) (import-outside-toplevel)
models\reservation.py:100:8: C0415: Import outside toplevel (models.hotel.Hotel) (import-outside-toplevel)
************* Module tests.test_customer
tests\test_customer.py:21:21: W0212: Access to a protected member _load_customers of a client class (protected-access)
tests\test_customer.py:28:21: W0212: Access to a protected member _load_customers of a client class (protected-access)
tests\test_customer.py:34:12: W0212: Access to a protected member _save_customers of a client class (protected-access)
************* Module tests.test_hotel
tests\test_hotel.py:21:21: W0212: Access to a protected member _load_hotels of a client class (protected-access)
tests\test_hotel.py:28:21: W0212: Access to a protected member _load_hotels of a client class (protected-access)
tests\test_hotel.py:34:12: W0212: Access to a protected member _save_hotels of a client class (protected-access)
************* Module tests.test_reservation
tests\test_reservation.py:23:21: W0212: Access to a protected member _load_reservations of a client class (protected-access)
tests\test_reservation.py:30:21: W0212: Access to a protected member _load_reservations of a client class (protected-access)
tests\test_reservation.py:36:12: W0212: Access to a protected member _save_reservations of a client class (protected-access)
************* Module main
main.py:1:0: R0801: Similar lines in 2 files
==models.customer:[10:17]
==models.hotel:[10:17]
    os.makedirs(os.path.dirname(DATA_FILE), exist_ok=True)
    if not os.path.exists(DATA_FILE):
        return {}
    try:
        with open(DATA_FILE, 'r', encoding='utf-8') as f:
            return json.load(f)
    except (json.JSONDecodeError, IOError) as e: (duplicate-code)
main.py:1:0: R0801: Similar lines in 2 files
==models.customer:[18:28]
==models.hotel:[18:28]
        return {}


def _save_hotels(data):
    """Persist hotels dict to JSON file."""
    os.makedirs(os.path.dirname(DATA_FILE), exist_ok=True)
    try:
        with open(DATA_FILE, 'w', encoding='utf-8') as f:
            json.dump(data, f, indent=4)
    except IOError as e: (duplicate-code)

------------------------------------
Your code has been rated at 9.70/10
```

End Result

```
-------------------------------------------------------------------
Your code has been rated at 9.14/10 (previous run: 9.14/10, +0.00)

(base) PS C:\Users\delrivero.m\a01362368_NumerodeActividadA6.2> python -m pylint models/ tests/ main.py

-------------------------------------------------------------------
```

Evidencias Cobertura de líneas por clase

```
OK
(base) PS C:\Users\delrivero.m\a01362368_NumerodeActividadA6.2> python -m coverage report -m
Name                        Stmts   Miss  Cover   Missing
-----------------------------------------------------------
models\__init__.py              0      0   100%
models\customer.py             61      0   100%
models\hotel.py                95      0   100%
models\persistence.py          19      0   100%
models\reservation.py          57      0   100%
tests\__init__.py               0      0   100%
tests\base.py                  16      1    94%   20
tests\test_customer.py         49      1    98%   79
tests\test_hotel.py           101      2    98%   118, 162
tests\test_reservation.py      81      2    98%   53, 123
-----------------------------------------------------------
TOTAL                         479      6    99%
(base) PS C:\Users\delrivero.m\a01362368_NumerodeActividadA6.2>
```